

# Prakrititz Borah

Bangalore, India

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## Education

International Institute of Information Technology Bangalore  
*Integrated MTech (B.Tech + M.Tech) in Computer Science and Engineering*

Jul 2023 – Jul 2028  
CGPA: 3.35/4.00

- Teaching Assistant (Digital Design, Data Structures and Algorithms): Mentored and tutored 250+ students.

## Work Experience

Research Work – Samsung Lab, IIIT-Bangalore, under Dr. Sakshi Arora

Aug 2025 – Dec 2025

- Collected and curated a **real-world wearable ECG dataset** from **51 subjects** spanning **20 hours** of recordings (Polar H10); proposed an end-to-end **stress detection pipeline** including signal reconstruction, bandpass filtering, and HRV segmentation.
- Applied **GA-based feature selection** to reduce **43→22 HRV features**, achieving **74% accuracy** at **99.9% lower compute**, enabling deployment on **edge-constrained hardware**.
- Designed **CTBFNet** (CNN–Transformer–BiLSTM) with **HRV-guided cross-attention**, reaching **79% accuracy**; analyzed accuracy–efficiency trade-offs for **edge deployment** viability.

## Achievements

- Amazon Sambhav Hackathon (2024): 1st Among 100k+ teams**; invited to **Amazon Seattle** for a leadership gathering.
- Best Paper Candidate – BIOSTEC 2026 (Marbella, Spain)**: Nominated for best paper among international submissions.
- Goal Guard (SaaS Productivity App)**: Scaled to **10K+ downloads across 20+ countries** on Google Play.
- Max Expert on Codeforces** with peak rating of **1641**. 1000+ problems solved across different platforms.
- Rank 61 in CodeQuest 2.0** among **3,000+ participants** in a competitive programming contest organized by IIIT Kottayam.

## Projects

RelayBrain | *Node.js, MCP, AI Agents, Open-Source* | [link](#)

May 2026 – Jun 2026

- Developed an **open-source npm package** with a **dedicated CLI** syncing shared context layer across Claude Code, Cursor, Copilot, and Codex, **version-tracking** agent decisions and task history across tool switches.
- Designed **stop hooks** that parse session transcripts into structured **knowledge snapshots** compiled into a shared memory.
- Exposed an **MCP server** over stdio for structured agent read/write access with a zero-database, file-only design.

Token-Pruning – RAG Prompt Compression | *PyTorch, Hugging Face, LangChain* | [link](#)

Feb 2026 – Apr 2026

- Fine-tuned **CodeT5-B(220M)** for code-specific LLM context compression on **471K samples** across 9 compression ratios.
- Refined a **RAG pipeline** (BM25, CodeLlama-13B) for fixing code bugs, evaluated on **Microsoft's CodeXGLUE** using **AST-based type-aware priority ranking**.
- Achieved a **41% token reduction** while maintaining an **80.36 CodeBLEU score**. Deployed to Hugging Face.

Mini-Vercel – Self-Hosted PaaS | *Node.js, Docker, Caddy, SQLite* | [link](#)

Jan 2026 – Feb 2026

- Built a self-hosted PaaS that auto-generates and deploys Dockerfiles achieving **zero-downtime blue-green deploys in ~25 seconds**; supports **framework detection** for Next.js, Vite, CRA, and Node.js.
- Implemented a **GitHub webhook-triggered FIFO build queue** with rollbacks, PR previews, and custom domains via Caddy-managed **automatic HTTPS** (local CA + Let's Encrypt).
- Secured secrets with **AES-256 encrypted** GitHub PAT storage and **bcrypt-hashed** session auth; **SQLite** stores project metadata, env vars, and deployment history.

GoalGuard – Productivity App | *Kotlin* | [link](#)

Jul 2025 – Dec 2025

- 1500+ daily active users**; integrated **Google Play Billing** for secure subscription validation and feature gating.
- Executed **app and site blocking** via **Android system APIs** with lifecycle-aware services, persisting state across OS restarts.
- Implemented **real-time UI sync** between blocking engine and foreground service; added haptic feedback for live status changes.

## Technical Skills

**Languages:** C++ | Python | Java | Kotlin | JavaScript | C | SQL

**Technologies & Frameworks:** PyTorch | Hugging Face | Transformers | LangChain | Flask | Node.js | SpringBoot | Android SDK

**DevOps:** Docker | Caddy | Git | MCP | Kubernetes

**Relevant Coursework:** Data Structures and Algorithms | Database and Management Systems | Computer Networks | Operating Systems | Object Oriented Programming | Multi Objective ML | NLP with Deep Learning | Algorithmic Thinking | Number Theory.

## Publications

Stress Detection from Consumer-Grade ECG Device Using GA Optimization

*19th International Joint Conference on Biomedical Engineering Systems and Technologies* | [link](#)

Mar 2026  
Marbella, Spain

## Extracurricular

**ICP Crewsphere IIITB Chapter:** Established the campus blockchain club, organized hackathons and technical workshops.

**Zense (Tech Club), IIIT Bangalore:** Conducted workshops, developed the college's official fest website, and organized tech events.

**Film Division, IIIT Bangalore:** Organized competitions and produced two award-winning films.